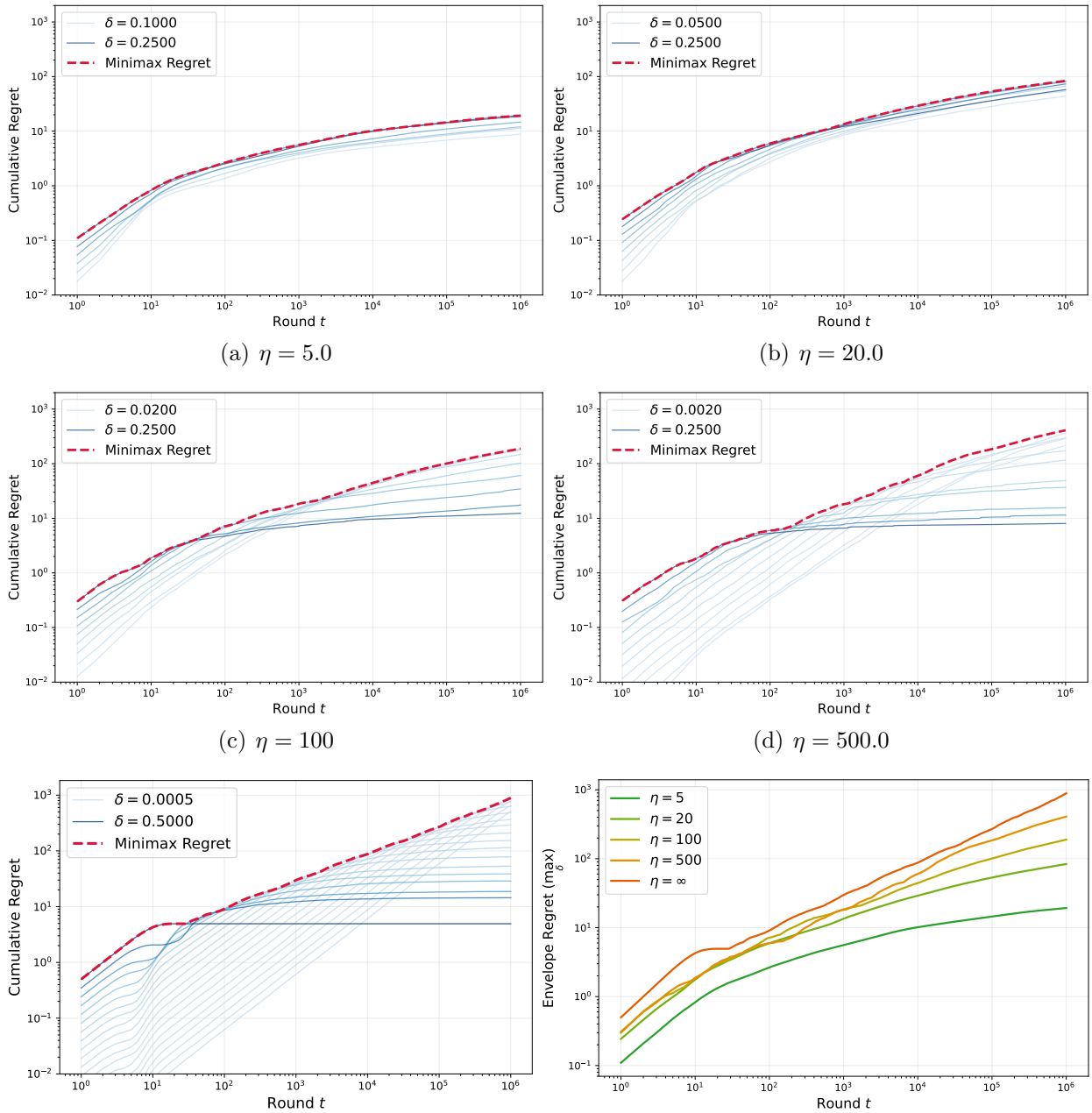
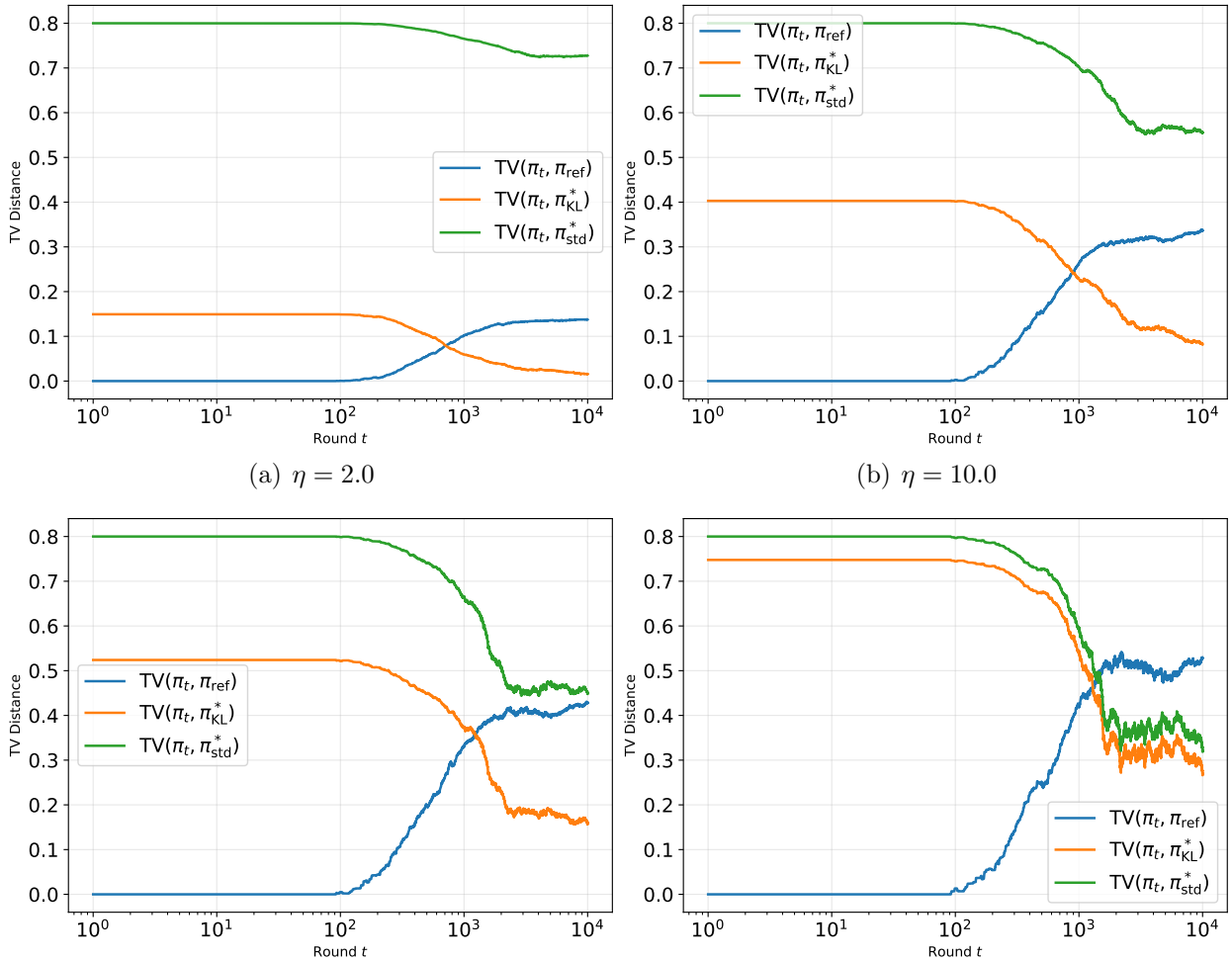


(a) $\eta = 5.0$ (b) $\eta = 20.0$

(c) $\eta = 100$ (d) $\eta = 500.0$

(e) $\eta = \infty$ (f) Comparison over η

Figure 1: The regret curves of KL-UCB for different values of η. In these experiments, we consider $\eta \in \{5, 20, 100, 500, \infty\}$, where $\eta = \infty$ corresponds to the standard MAB setting. We set $K = 4$ and $T = 10^6$. The bandit instances are defined by $\pi^{\mathrm{ref}} = \mathsf{Unif}(\mathcal{A})$ and rewards $r = [\delta, 2\delta, 0, 0]$, where δ is swept log-uniformly over an appropriate interval. For each fixed η, the regret corresponding to each instance (determined by δ) is plotted as a single blue solid line in Figures 1(a)–1(e). The **red bold dashed lines** represent the maximum regret over all instances, and therefore approximate the minimax regrets. For ease of comparison, all minimax regret curves are plotted together in Figure 1(f). **Both axes are shown on a log scale.** We observe that the minimax regret of the standard MAB exhibits the $\sqrt{T}$ rate when $t > 100$. In contrast, when $\eta < \infty$, the slope of the regret curve decreases continuously as t increases, with more pronounced decreases for smaller values of η. These results are consistent with the theorems proved in the paper.



(a) $\eta = 2.0$ (b) $\eta = 10.0$

(c) $\eta = 20.0$ (d) $\eta = 50.0$

Figure 2: Policy convergence in TV distance across training rounds $t \in [T]$ for varying regularization strengths $\eta \in \{2, 10, 20, 50\}$, with $K = 5$, $T = 10^4$ and $\pi^{\mathsf{ref}} = \mathsf{Unif}(\mathcal{A})$. Each subfigure plots the TV distance between the learned policy π_t and three comparison policies: the reference policy π_{ref}, the optimal policy π^*_{KL} for the KL-regularized objective, and the deterministic optimal policy π^*_{std} for the standard objective. For small η (high regularization), π_t remains close to π_{ref} and converges toward π^*_{KL}, consistent with the KL penalty dominating the objective. As η increases, the regularization weakens, and the policy moves further away from π_{ref}, instead moving close π^*_{std}, thereby recovering the unregularized optimum. This behavior illustrates how η interpolates between the two regimes, complementing our theoretical guarantees on policy convergence.